

Program Guide — Golf Cart Diagnostic Technician Training Program

Version: 1.1 · July 2026 · Confidential Training Material

Contact hours: 40 · **Sessions:** 20 · **Recommended hands-on ratio:** ≥60%

1. Purpose & Graduate Outcomes

This program equips technicians to accurately diagnose modern golf carts — electric (flooded lead-acid and lithium) and gas-powered — using safety-first habits and a systematic 7-step diagnostic method.

Upon successful completion, graduates will be able to:

1. Apply LOTO, PPE selection, and high-current / chemical hazard controls on golf cart systems.
 2. Use DMM, load tester, hydrometer/refractometer, and basic diagnostic tools correctly and safely.
 3. Diagnose battery, charger, motor, controller, throttle, solenoid, and wiring faults on Club Car, EZ-GO, Yamaha, and common aftermarket systems.
 4. Read OEM wiring diagrams and apply the 7-step diagnostic methodology to real symptoms.
 5. Inspect brakes, suspension, steering, and drivetrain for common failure modes.
 6. Differentiate lead-acid vs lithium diagnostics and explain BMS protective functions.
 7. Execute preventive maintenance and document findings to professional shop standards.
-

2. Learning Objectives Map (Week → Objective → Assessment)

ID	Learning Objective	Weeks	Primary Assessment
LO-01	Select and use correct PPE; perform LOTO; verify zero energy	1	Safety quiz; LOTO sign-off
LO-02	Apply Ohm's Law; measure V/R/continuity with DMM	1	Multimeter competency checklist
LO-03	Inspect and diagnose flooded lead-acid banks (OCV, SG, load)	2	Battery test log accuracy
LO-04	Troubleshoot chargers; explain charge stages	3	Charger troubleshooting worksheet
LO-05	Perform basic lithium/BMS diagnostics; contrast with lead-acid	3	Comparison quiz; lithium demo sheet
LO-06	Inspect/test DC motors (brushes, commutator, resistance, insulation)	4	Motor inspection report
LO-07	Interpret controller fault codes; test throttle & solenoid	5	Fault-code exercise; test sheets
LO-08	Navigate OEM diagrams; locate opens/shorts/high resistance	6	Diagram quiz; planted-fault lab
LO-09	Apply 7-step process; document diagnosis professionally	7, 10	Process checklist; case study; final practical
LO-10	Diagnose mechanical systems via inspection & road test	8	Mechanical checklist + road-test notes
LO-11	Perform basic gas-cart checks (compression, spark, fuel)	9	Gas diagnostic checklist
LO-12	Design/execute fleet PM schedules and checklists	10	PM plan exercise; written final

3. Assessment System & Rubrics

3.1 Grading Weights (suggested)

Component	Weight	Passing
Weekly quizzes (average)	15%	≥70% each or average ≥70%
Lab worksheets / competency sign-offs	30%	All critical safety sign-offs = Pass
Week 7 & case study documentation	15%	Rubric ≥3.0 / 4.0
Final practical skills assessment	25%	Rubric ≥3.0 / 4.0 on all critical items
Written final exam (50 questions)	15%	≥70% (35/50)

Certificate requirement: Pass all critical safety competencies; final practical ≥ 3.0 ; written $\geq 70\%$; attendance $\geq 85\%$ (or makeup labs completed).

3.2 Performance Scale (used on practical rubrics)

Score	Label	Descriptor
4	Proficient	Correct, safe, complete with minimal cues; professional documentation
3	Competent	Correct outcome; minor omissions or one non-critical cue
2	Developing	Incomplete process; multiple cues; safety prompt required once
1	Unsatisfactory	Incorrect diagnosis, unsafe practice, or critical step skipped
0	Not attempted / Unsafe stop	Instructor stops work for safety

Critical items (must score ≥ 3): PPE, LOTO/zero-energy verify, meter category/probe discipline, no jewelry, no interlock bypass.

3.3 Documentation Rubric (reports & case studies)

Criterion	4	3	2	1
Symptom verification	Clear, specific, verified	Mostly clear	Vague	Missing
Test data	Complete, units, conditions noted	Minor gaps	Incomplete	None / invented
Reasoning chain	Logical isolation path	Mostly logical	Jumps to parts	Parts-swap mentality
Recommendation	Actionable & justified	Adequate	Weak	Incorrect
Safety notes	Explicit	Present	Implied only	Ignored

4. Suggested Class Size & Instructor Ratio

Parameter	Ideal	Acceptable	Notes
Students	6-8	4-10	Above 10 splits into two lab groups
Instructor : student	1:8	1:10 max	Add assistant for >8 or multi-cart faults
Carts available	4-6	3 minimum	Mix 36V/48V LA, ≥ 1 lithium, 1-2 gas
DMM per students	1:2	1:3 max	Prefer one meter per student when possible
Lab team size	2	2-3	Pair work for Weeks 2-9; solo for final practical

5. Program Format & Weekly Rhythm

- **Session 1 (Theory/Demo, 2 hrs):** Concepts, demos, whiteboard/schematics, short knowledge checks.
 - **Session 2 (Lab, 2 hrs):** Stations, planted faults, worksheets, debrief.
 - **Every session:** Opening safety (2-5 min), closing cleanup & parking (5-10 min).
 - **Homework (optional, 15-30 min):** Read handout for next week; complete short reflection on lab data.
-

6. Prerequisites & Target Audience

Prerequisites: Basic mechanical aptitude; ability to follow written procedures; valid driver's license for moving carts. Prior small-engine or 12/24/48V experience helpful but not required.

Audience: New golf cart techs; maintenance staff at courses/resorts/HOAs/fleets; experienced mechanics specializing; supervisors deepening diagnostic capability.

7. Required Equipment & Facilities (summary)

See also weekly instructor prep lists and Appendix-equivalent tool list in Week 1 materials.

Must have: 4-6 carts (mixed), CAT III DMMs, battery load tester (100A+), hydrometer/refractometer + temp chart, hand tools, torque wrenches, crimpers, PPE (glasses, Class 00 gloves, leather over-gloves, face shields), Class C/ABC extinguisher, spill kit, eyewash, ventilation for charging, OEM manuals (Club Car, EZ-GO, Yamaha), Curtis fault-code reference (print or <https://faultcodes.curtisinstruments.com/>).

Nice to have: Curtis/Alltrax programmer, megger, training boards, large schematic display, lithium OEM app/tooling.

8. Scaling Notes

Situation	Adaptation
Small class (≤4)	More one-on-one coaching; each student gets full cart ownership for case study
Large class (10-12)	Split Session 2 into A/B rotations; second instructor or senior tech as proctor
Fleet-only (one brand)	Keep multi-brand theory; deepen labs on fleet OEM diagrams
Lithium-heavy shop	Expand Week 3; add BMS app stations; still teach lead-acid (legacy fleets)
No gas carts available	Use engine stand + video for Week 9; keep written assessment; mark practical “demo observed”
Compressed delivery	Not recommended below 40 hrs; if forced, merge Week 8 mechanical into Week 7 debrief and Week 10
Online theory hybrid	Theory may be remote; all LOTO, battery, high-current, gas labs must be in-person

9. Customization Tips

1. **Insert shop SOPs** (torque specs, work-order software screenshots) into Week 7 documentation block.
2. **Plant faults that match your common comebacks** (high-resistance cables, weak solenoids, mis-set HPD).
3. **Localize PM schedules** to your climate (Coastal humidity → corrosion emphasis; cold climates → lithium charge-temp limits).
4. **Career pathways:** Tailor Week 10 to dealer OEM paths vs independent shop vs internal fleet promotion.
5. **Do not** teach mobile trip charges or discontinued product lines as current practice unless historical context is labeled “legacy.”
6. Update Curtis/Alltrax model list annually; always require model-year-correct diagrams.

10. The 7-Step Diagnostic Process (Program Spine)

Taught in Week 7; practiced Weeks 2–10; assessed on final practical.

1. **Verify the complaint** — Reproduce or clearly define the symptom.
2. **Gather information & history** — Interview, service records, recent work, environment.
3. **Visual & safety inspection** — PPE, damage, corrosion, leaks, loose hardware.
4. **Isolate systems** — Power → control → propulsion → mechanical; start with batteries/connections.
5. **Test components** — Measure, don’t guess; record values with units.
6. **Perform repair** — Correct root cause; torque/specs; no safety bypasses.

7. **Verify repair & road test** — Confirm fix under load; document.

Shop mantra: Diagnose before replace.

11. Attendance, Makeup & Conduct

- Missed theory: complete reading + quiz makeup before next lab.
 - Missed lab: scheduled makeup with instructor before Week 10 finals.
 - Jewelry, horseplay, or intentional interlock defeat → immediate stop + coaching; repeat → dismissal from session.
-

12. Certificate Title

Golf Cart Diagnostic Technician — Certificate of Completion

(See `final_assessment/certificate_template.md`)

13. Versioning & Continuous Improvement

Field	Value
Curriculum base	High-level 10-week program (v1.0, July 2026)
Package version	1.1 (expanded delivery package)
Review cadence	Annual or when major OEM/controller families change
Feedback loop	Week 10 student survey + instructor after-action notes → update <code>ITERATION_LOG</code>

14. File Cross-Reference

Need	Go to
Timed agendas	<code>weeks/week_XX_*.md</code>
Printable lab forms	<code>labs/</code>
Student pocket references	<code>handouts/</code>
Finals	<code>final_assessment/</code>
Self-review status	<code>ITERATION_LOG.md</code>